# Mining Software Repositories

Organization



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# Administration

- Lecture
  - Monday, 10:15-11:45 o'clock
  - Room: Building HK 28 – SR 010
- Exercises
  - Wednesday, 10:15-11:45 o'clock
  - Room: Building HK 14b – SR 017
- Exam date 1:
  - Wednesday, Feb 10th, 14:00-16:00 o'clock
- Exam date 2:
  - Monday, Mar 22nd, 10:00-12:00 o'clock
- The exam dates are preliminary.
  - Please check the central document regarding registration date: https://www.uni-passau.de/en/examinations-office/semester-related-information
  - You are not allowed to participate in the exam if you fail to register in time!

# Exercise

- Application of Concepts from the Lecture

- Weekly exercise
  - You can freely decide to which slot you go

- Combination of theoretical exercises and practical application development
  - Can be solved alone or in groups

- Exact format will depend on number of (active) participants during the exercise
  - If there are only few participants:
    - Students/group will get individual feedback for their practical solutions (“Testat”)
    - Theory tasks will afterwards be discussed with the whole group
  - If there are too many participants for the above:
    - Everything will be discussed as a group based on the student solutions

# Exercises: What they are and are not

## The exercises are …

- optional in the sense that you do not automatically fail if you do not do the exercise
- for giving feedback for your solutions and give you hints on how to solve stuff
- an opportunity to better learn the concepts and work on your own solutions
- for learning about the contents of the lecture

## The exercises are not …

- optional in the sense that you can expect a good grade (or even passing) if you do not solve the exercises
- a basic programming tutorial or the place where you start working on the solutions
- a second lecture in which we present or give solutions
- for discussing administrative issues

# Communication and materials

- Everything runs via StudIP
  - Lecture slides
  - Exercise sheets
  - Other announcements from the lecturers
  - Literature hints

- Make sure you are enrolled in both the lecture and the exercise

# Learning goals

- Know the role of MSR for software engineering tools and evidence-based software engineering
- Understand how to collect data from repositories and how this data can be enriched
- Understand how to design MSR studies and tools
- Be able to apply all of the above

# Prerequisites

- Some machine-learning knowledge required
  - Foundations in data science
    - Data cleaning, feature engineering, statistics, machine learning
  - Software engineering incl. processes and quality assurance
    - E.g., version control, issue tracking, quality assurance approaches
  - Good programming knowledge
    - Most will be Python
    - Should already know common data science libraries like pandas, matplotlib and scikit-learn
    - Should be able to work with new libraries and tools just based on their documentation
  - Knowledge about working with and setting up databases

# Contents of the lecture

- MSR data collection
  - Data sources and collecting raw data
  - Enriching data with heuristics and manual validation
  - Software metrics
  - Developer social networks
- Empirical methods
  - Research paradigms
  - Sampling
  - Basic statistics and experiment designs
  - Basics of qualitative analysis
  - Threats to validity
  - …
- Applications of MSR
  - Understanding defects
  - Studying social aspects of software engineering
  - Visualizing code and its history
  - …

Questions?

